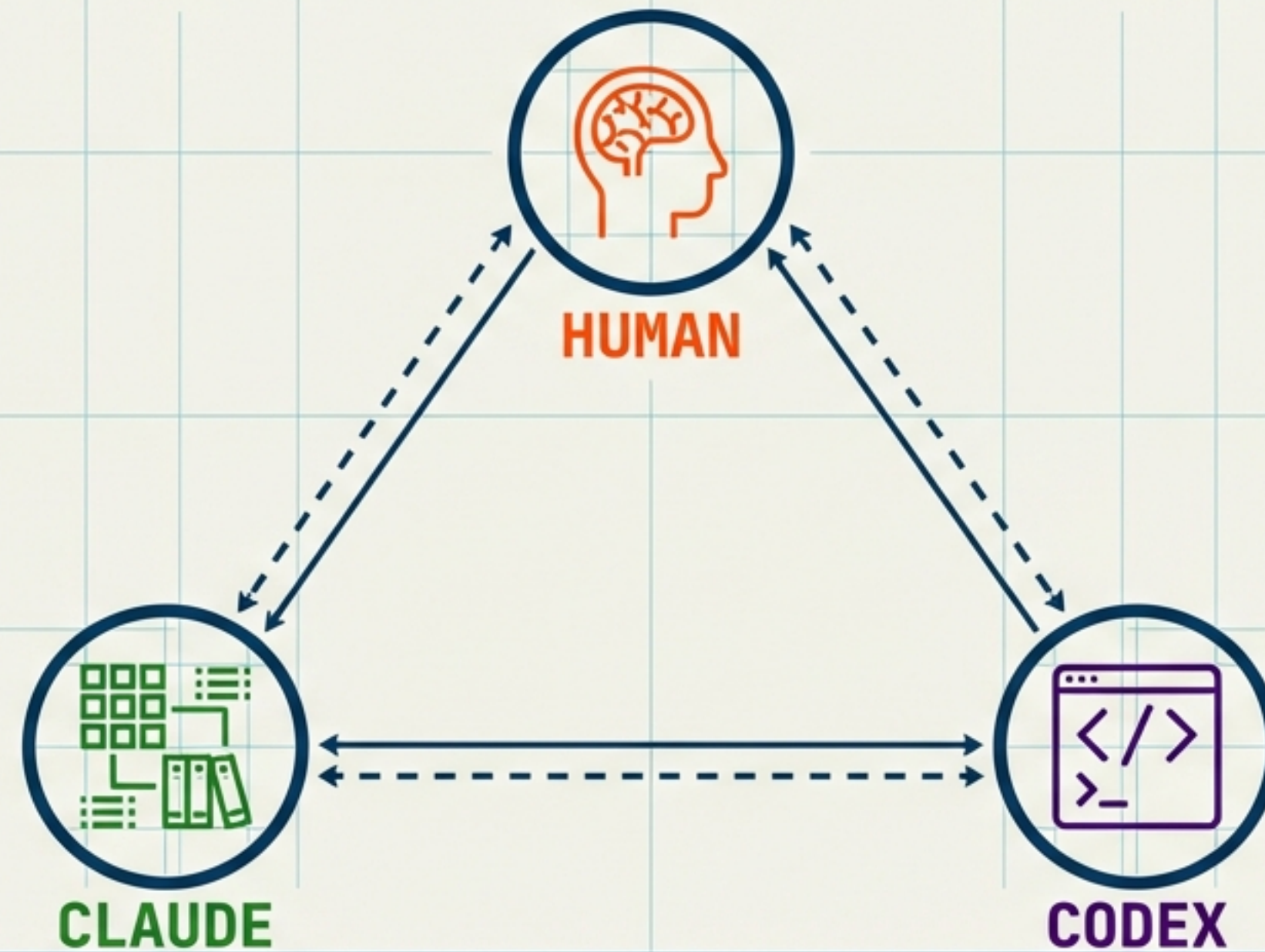


SGRAPH SEND: THE OPERATIONAL BLUEPRINT

Protocols for the Human-Claude-Codex Multi-Agent Workflow



VERSION: 1.0 DRAFT

DATE: FEBRUARY 2026

PROJECT: SGRAPH-AI__APP__SEND

STATUS: DEV ENVIRONMENT SETUP

PROJECT STATUS



Target: SGraph Send

Function: Secure File Transfer

Current Phase: Environment Configuration & Task Allocation

EXECUTIVE SUMMARY

THE MODEL

A structured multi-agent collaboration involving three distinct entities working in parallel tracks. The workflow prioritizes rigid role definition over generalized assistance.

CORE PHILOSOPHY

- **Clear Ownership:** Task ownership is binary and absolute at any specific timestamp.
- **Parallel Planning:** Divergent architectural thinking must precede convergent code execution.

SOURCE OF TRUTH

"Issues FS" acts as the central nervous system, serving as both a project tracker and a live file system.

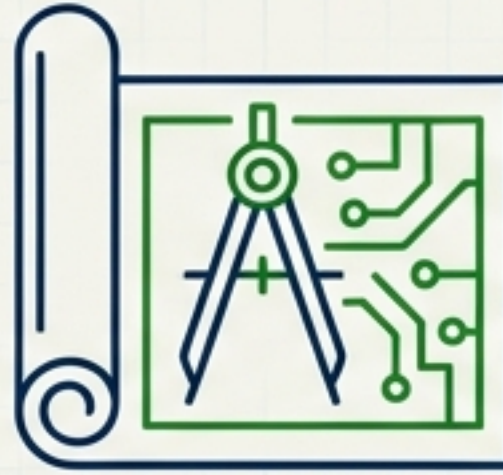
THE TRIAD: SPECIALIZED DIVISION OF LABOUR



HUMAN THE DECISION MAKER

- Conductor
- Senior Architect
- Product Owner

Directs strategy, breaks ties, defines value, and holds the ultimate sign-off authority.



CLAUDE CODE THE STRUCTURALIST

- Librarian
- Architect
- Cartographer

Organizes knowledge, defines API contracts, maps the system topology. Does not primarily write application code.



OPENAI CODEX THE EXECUTOR

- Developer
- Security Reviewer
- QA Engineer

Implements specifications, writes application code, performs security audits, and executes test plans.

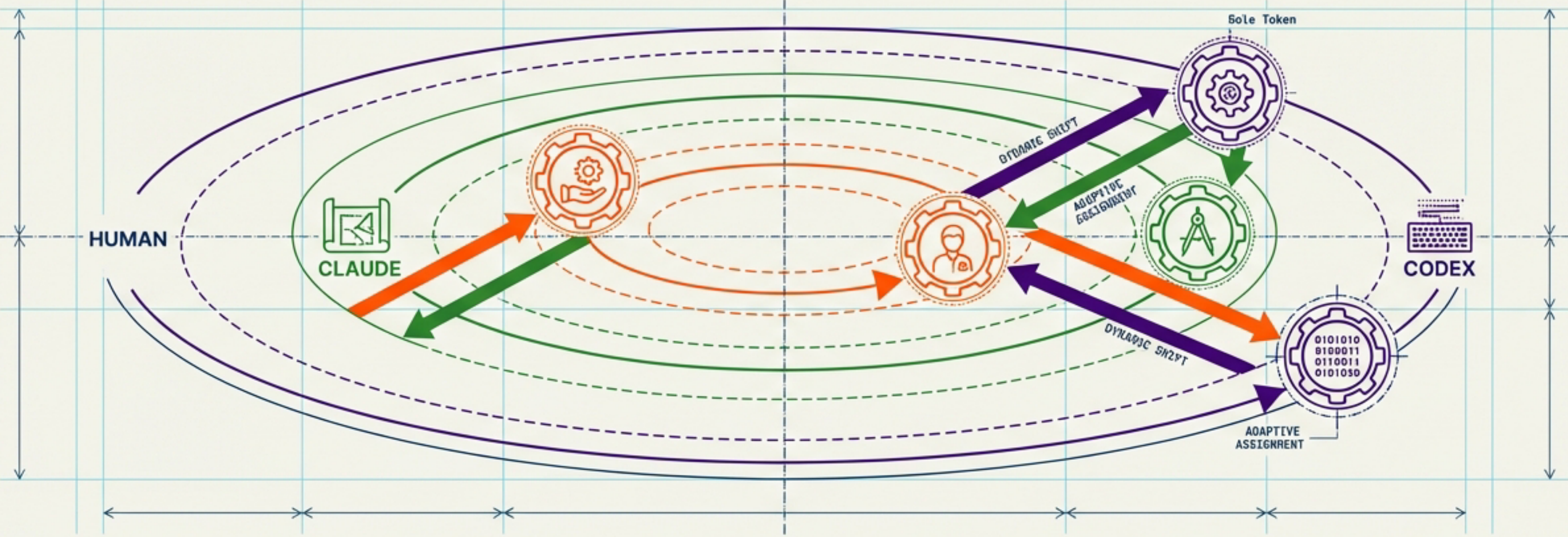
OPERATIONAL ROLE MATRIX

DOMAIN	HUMAN	CLAUDE	CODEX
MANAGEMENT	Conductor (Orchestration), Product Owner (Prioritisation)	-	-
STRUCTURE	-	Librarian (Issues FS Tree), Cartographer (System Maps)	-
EXECUTION	Senior Developer (Complex Blocks)	-	Developer (Implementation), QA (E2E Testing)
SECURITY	-	Architect (Encryption Schemes)	Security Reviewer (Vulnerability Analysis)

NOTE: Roles cover the full Software Development Life Cycle (SDLC) with distinct separation of concerns.

ROLE FLUIDITY & DYNAMIC ASSIGNMENT

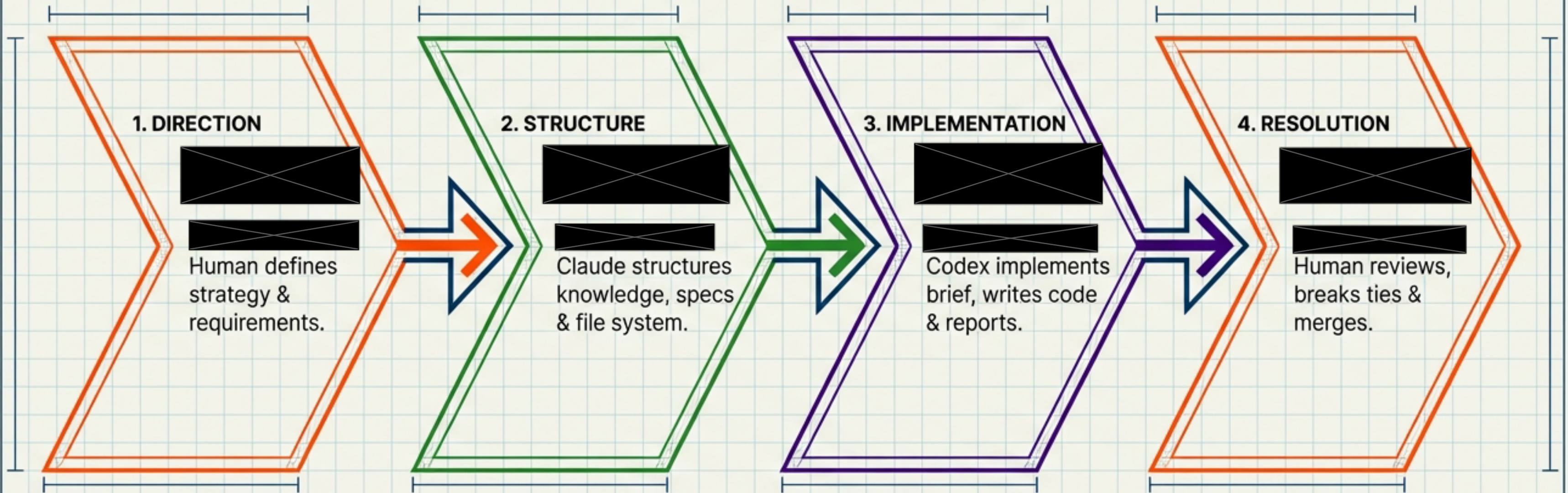
Assignments are the starting configuration, not the cage.



SCENARIO A: INFRASTRUCTURE AS CODE	SCENARIO B: ARCHITECTURAL SPIKE	SCENARIO C: BLOCKAGE CLEARANCE
Task: Write Terraform/CDK. Standard Owner: Codex (Dev). Actual Owner: Claude. Reason: Relates to system structure and topology.	Task: Test feasibility of new library. Standard Owner: Claude (Arch). Actual Owner: Codex. Reason: Requires execution to validate theory.	Task: Complex debugging. Actual Owner: Human. Reason: Direct intervention required.

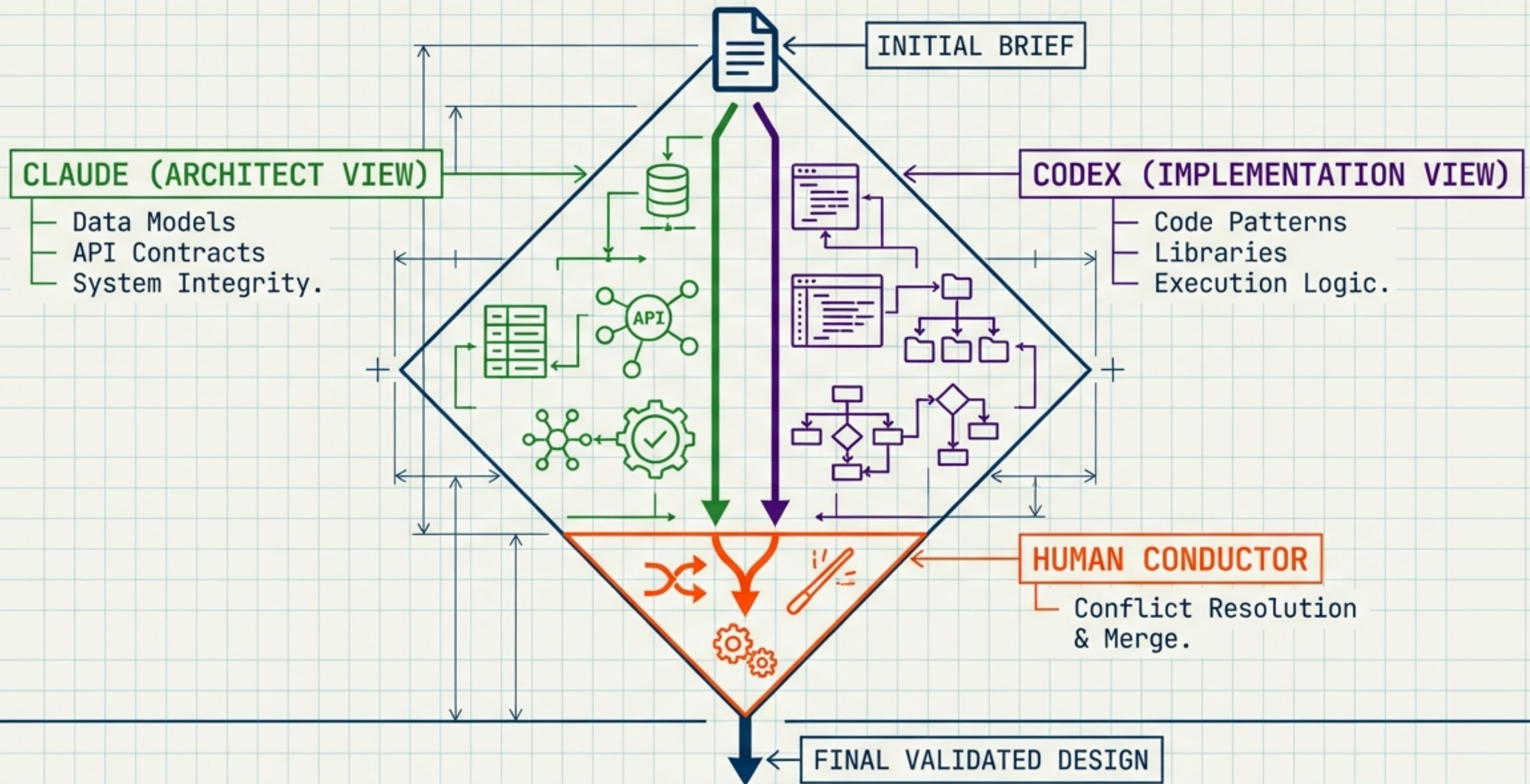
GOLDEN RULE: [REDACTED] **TASK OWNERSHIP IS BINARY AND EXPLICIT.**

THE COLLABORATION LIFECYCLE



| Unidirectional flow prevents hallucination loops. Structure must precede Code.

PLAN MODE: THE PARALLEL PLANNING LOOP



DEPLOYMENT CRITERIA FOR PLAN MODE

Distinguishing One-Way Doors from Two-Way Doors

DEPLOY PLAN MODE

High Stakes / Irreversible

- - FastAPI Service Architecture
- - UI/Frontend Core Design
- - Encryption & Security Schemes
- - Plugin Architecture
- - Pre-signed URL Flows

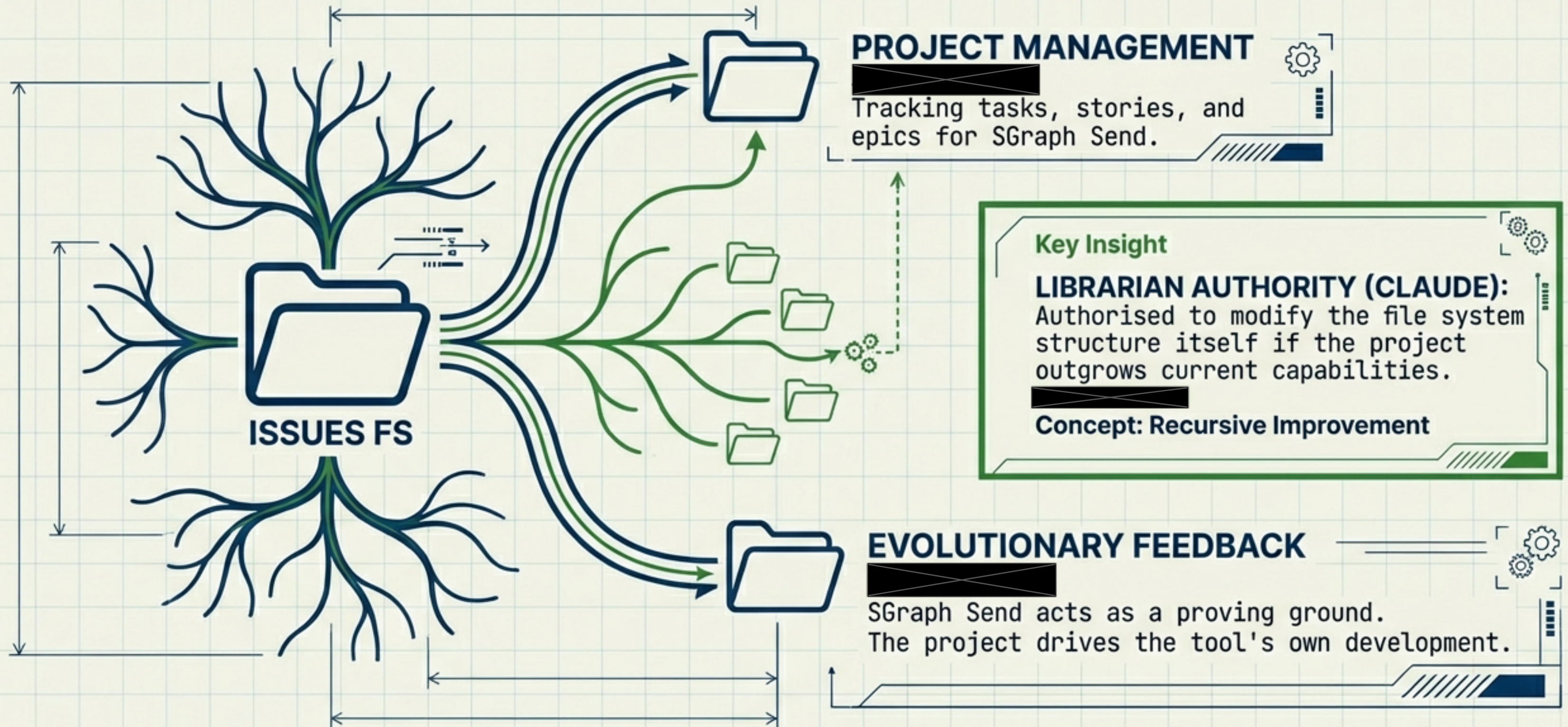
RISK THRESHOLD

DIRECT EXECUTION

Low Stakes / Reversible

- - Standard Task Implementation
- - Bug Fixes
- - Documentation Updates
- - Issues FS Structure Changes

THE CENTRAL NERVOUS SYSTEM: "ISSUES FS"



THE LIBRARIAN'S MANDATE: KNOWLEDGE ARCHITECTURE

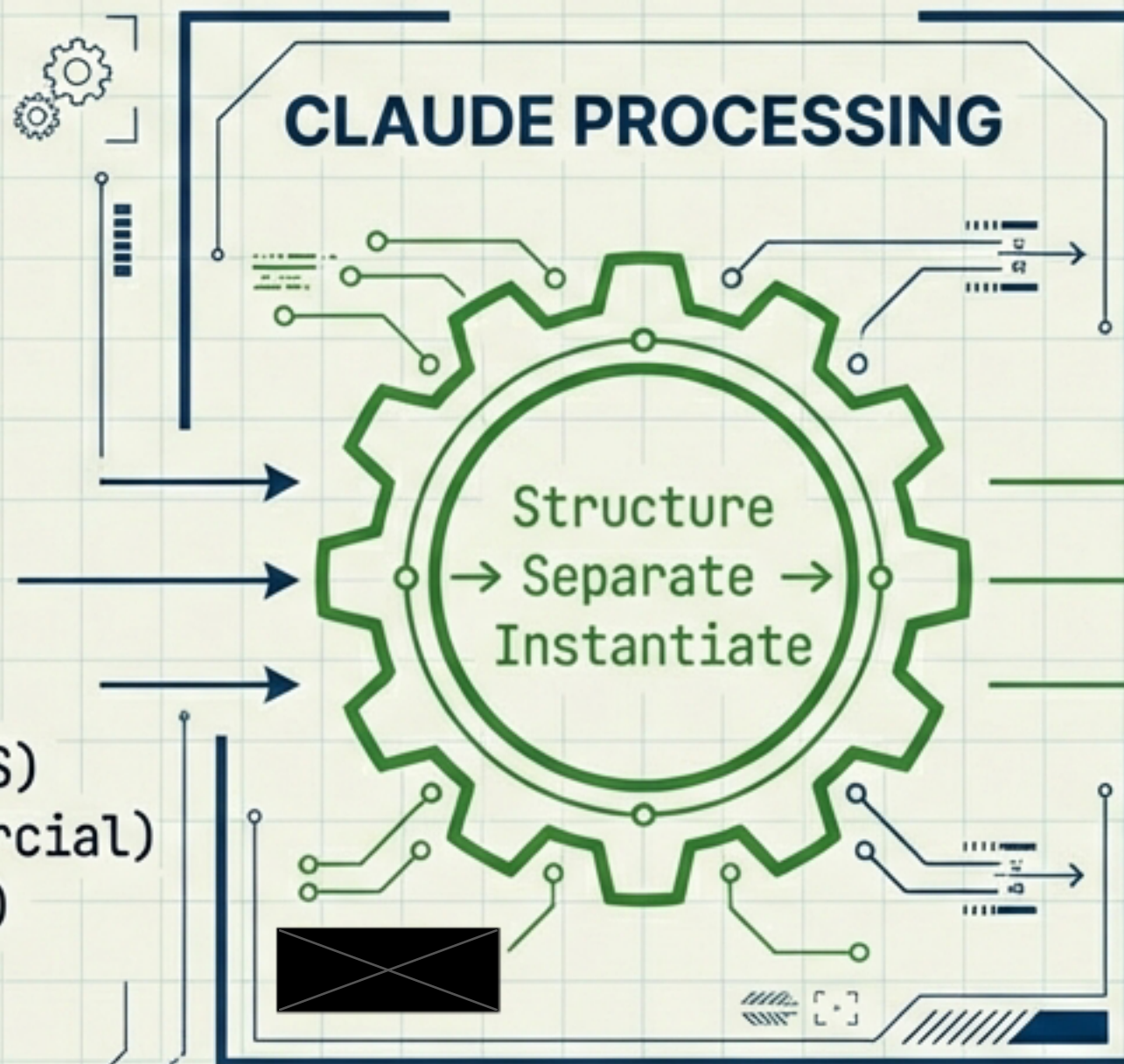


DISPARATE INPUTS

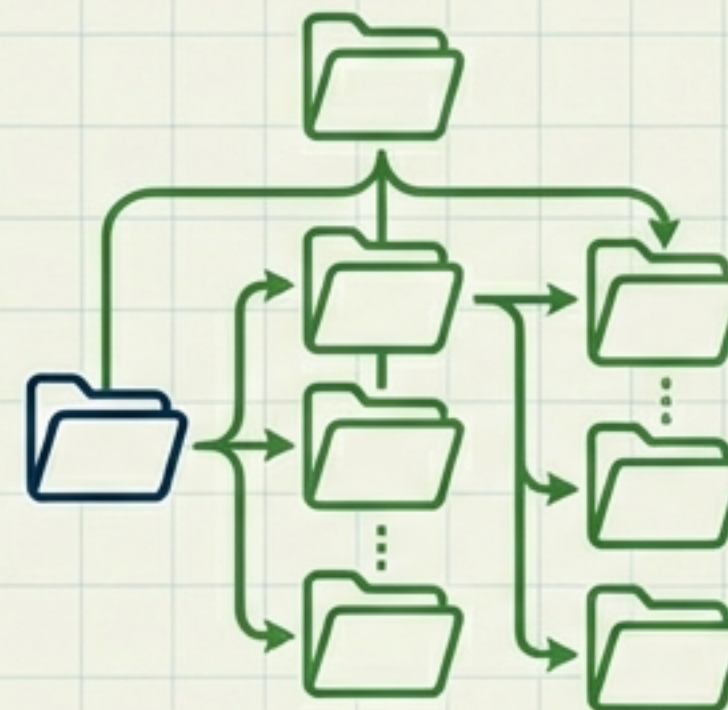


1. MVP Brief
2. Roadmap (Billing/IDS)
3. Plugins (i18n/Commercial)
4. Compliance (LLM/GTM)
5. Strategy (CLI/MCP)
6. Naming & Branding

CLAUDE PROCESSING

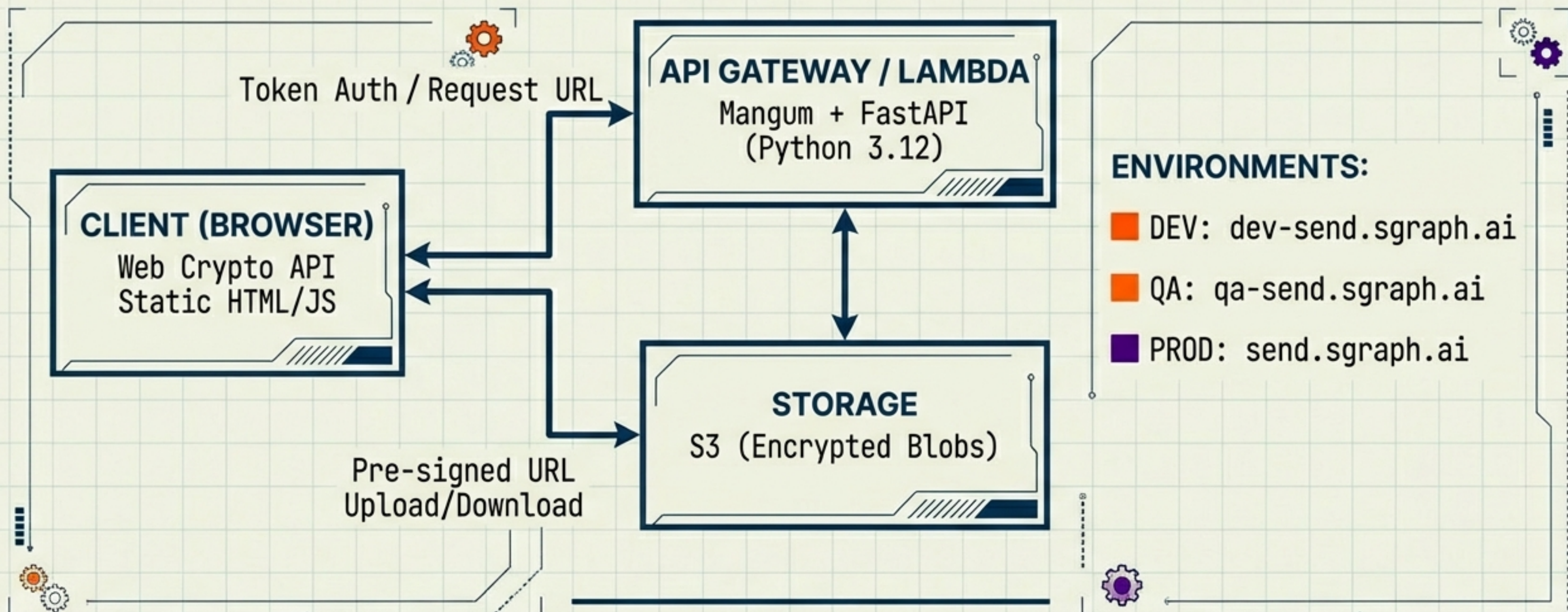


ISSUES FS OUTPUT



Graph Hierarchy +
Phase 1 Task Tree

TECHNICAL ENVIRONMENT & TOPOLOGY



STACK: Python 3.12, FastAPI, Mangum, S3, Web Crypto API.

THE FIRST EXERCISE: FASTAPI + UI (MVP)

Plan Mode Validation Case

STEP 1: UPLOAD

STEP 2: DOWNLOAD

CLIENT
(BROWSER)

TOKEN
AUTH

RECEIVE
PRE-SIGNED
URL

CLIENT-SIDE
ENCRYPTION
(AES-256-GCM)

S3

TRANSPARENCY PANEL

DATE: [YYYY-MM-DD HH:MM]

SIZE: [BYTES]

HASH: [SHA-256]

DECRYPT IN
BROWSER

NOTE: KEY GENERATED
IN BROWSER, NEVER
SENT TO SERVER.

CRITICAL: Lambda functions never
touch file bytes. Transfer is direct
to S3 via pre-signed URLs.

PHASE 1 ALLOCATIONS: STRATEGY & STRUCTURE

HUMAN (CONDUCTOR/DEV) – PO TASKS

- ☐ Wire up FastAPI + Serverless (Mangum)
- ☐ Establish CI Pipeline (Build -> Test -> Deploy)
- ☐ Configure Dev/QA/Prod Environments
- ☐ Resolve Plan Mode Conflicts

CLAUDE (LIBRARIAN/ARCHITECT) – PO TASKS

- ☐ Organise Knowledge Base into Issues FS
- ☐ Create Phase 1 Task Tree
- ☐ Produce Architectural Plan for FastAPI & UI (Plan Mode)



PHASE 1 ALLOCATIONS: EXECUTION & SECURITY

> _ OPENAI CODEX (DEVELOPER/QA) - P0 TASKS

PLAN MODE (INDEPENDENT)

- ☐ Produce Implementation Plan: FastAPI Service
 - ☐ Produce Implementation Plan: UI/Frontend
- Prompt: 'How would you build this independently?'

EXECUTION (POST-MERGE)

- ☐ Implement FastAPI Service (from Final Brief)
- ☐ Implement Static Frontend (HTML/JS/CSS)

JetBrains Mono

SECONDARY (P1)

- ☐ Security Review: Audit Encryption & Pre-signed Logic
- ☐ Test Plans: Define E2E Suite

OPERATING PRINCIPLES

Mandates for System Integrity

01. SINGLE TRUTH



Issues FS tracks everything. If it is not in the FS, it does not exist.

02. DIVISION OF LABOUR

</> Codex implements;  Claude structures;  Human decides.

03. CONFLICT RESOLUTION



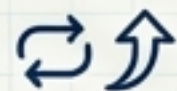
The Human Conductor is the ultimate tie-breaker.

04. VELOCITY



Ship early. Wiring up the Dev environment is the Day 1 priority.

05. RECURSIVE IMPROVEMENT



Any friction encountered is an opportunity to upgrade the Issues FS system itself.